

Project Initialization and Planning Phase

Date	01 Oct 2026
Team ID	SWTID-2026-8124
Project Name	ComicCraft - AI Comic Story Creator
Maximum Marks	5 Marks

Project Proposal (Proposed Solution) report

The proposal report aims to transform story creation using generative AI, letting users go from a simple idea to a fully illustrated comic. It tackles the time, skill and cost barriers of comic making. Key features include an AI story generator, automatic panel illustration and PDF export.

Project Overview	
Objective	To let anyone turn a short idea into a complete, illustrated comic story using generative AI, without needing writing or drawing skills, in a few minutes.
Scope	A web application that takes a story idea, genre and panel count; generates an outline (Gemini Flash), a detailed story with panel descriptions (Gemini Pro) and one image per panel (Stable Diffusion); and exports the finished comic as a PDF.
Problem Statement	
Description	Making a comic needs scripting, illustration and layout skills and usually takes days of effort. Students, teachers, writers and hobbyists often have story ideas but cannot turn them into visual stories quickly or affordably.
Impact	Solving this makes visual storytelling accessible to everyone, cuts creation time from days to minutes, lowers cost, and supports education, content creation and creative expression.
Proposed Solution	
Approach	A multi-stage generative AI pipeline: a fast model (Gemini Flash) drafts the outline, a stronger model (Gemini Pro) expands it into scenes, dialogue and image prompts, Stable Diffusion renders each panel, and FPDF assembles everything into a downloadable comic. FastAPI serves the workflow through a simple Jinja2 web interface.
Key Features	- AI-generated story outline from a single prompt (Gemini Flash). - Detailed story and panel-by-panel scripts (Gemini Pro). - Automatic panel image generation (Stable Diffusion via Diffusers). - Comic preview in the browser and one-click PDF export (FPDF).

Resource Requirements

Resource Type	Description	Specification/Allocation
Hardware		
Computing Resources	CPU/GPU specifications, number of cores	T4GPU(16GBVRAM)forStableDiffusion; multi-core CPU
Memory	RAM specifications	16 GB
Storage	Disk space for data, models, and logs	50GBSSD(model weights, generated images, PDFs)
Software		
Frameworks	Python frameworks	FastAPI, Uvicorn, Jinja2
Libraries	Additional libraries	google-generativeai, diffusers, torch, transformers, fpdf, Pillow, python-dotenv
Development Environment	IDE	VS Code, Jupyter Notebook / Google Colab
Data		
Data	Source, size, format	No training dataset; pre-trained Gemini(API) and StableDiffusion(HuggingFace) models. Inputs: text prompts. Outputs: PNG, PDF